

7. Hypothesis testing

7.3 Non-parametric tests

Non-parametric tests

- If the condition of normality of the data is not met, use **non-parametric tests**
- These do not require any specific distribution of the data
- Values of the data are **converted to ranks** (*remember the Spearman correlation!*)
- **Wilcoxon Rank Tests**
 - unpaired: ***Wilcoxon Rank Sum Test*** (a.k.a, Mann-Whitney U test)
 - paired : ***Wilcoxon signed rank test***

Wilcoxon Rank Sum Test Mann-Whitney U Test

- 2 samples with numerical values

$$X = \{x_1, x_2, \dots, x_{n_1}\} \quad Y = \{y_1, y_2, \dots, y_{n_2}\}$$

- Values are merged and ranked in increasing order

$$Z = X \cup Y$$

- R_1 is the sum of the ranks of the first probe (first probe is the one giving the smallest U)

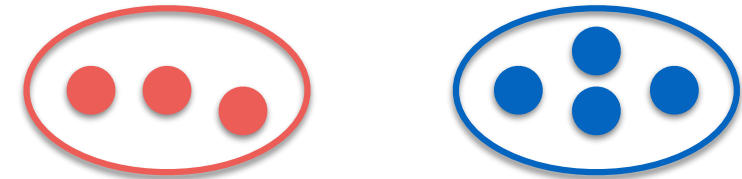
- Test statistics**

$$U = R_1 - \frac{n_1(n_1 + 1)}{2}$$

- H_0 :** $E(X) = E(Y)$ (2-sided test)

$$E(X) > E(Y) \quad (1\text{-sided test})$$

$$E(X) < E(Y)$$



Ranks

1



X_i

smallest

2



3



Y_i

4



5



$n_1 = 8$

6



$R_1 = 59$

.



.



$U = 23$

.

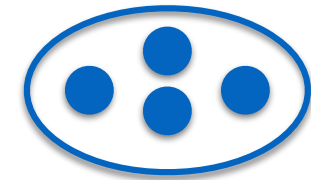
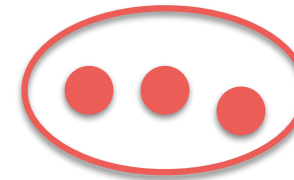


What is U if all reds are smaller than the blues?

largest

Wilcoxon Rank Sum Test Mann-Whitney U Test

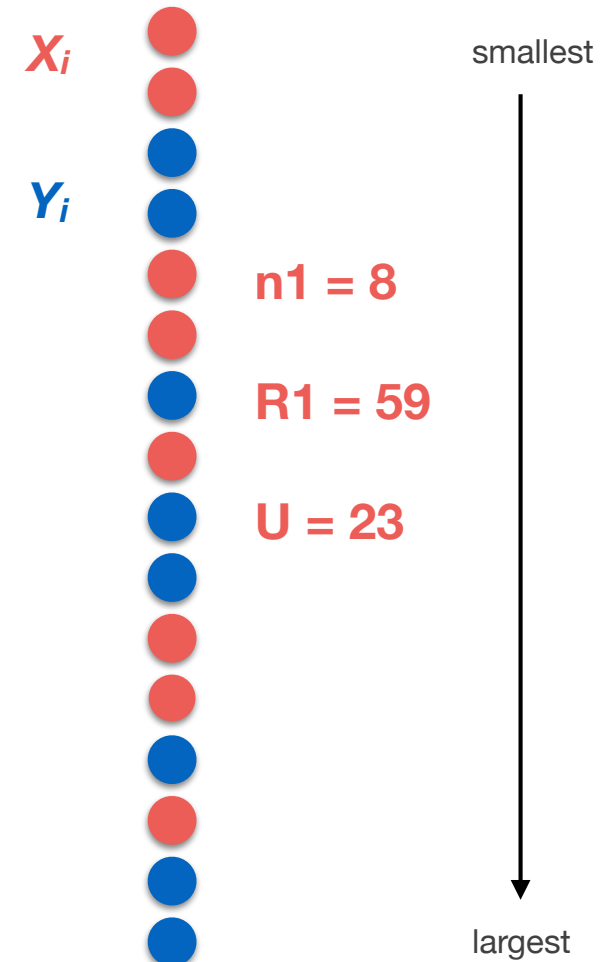
*Remember: the smaller U,
the more significant*



!! Values of α are for two-sided test !!

n_2	α	n_1												
		3	4	5	6	7	8	9	10	11	12	13	14	15
3	.05	--	0	0	1	1	2	2	3	3	4	4	5	5
	.01	--	0	0	0	0	0	0	0	0	1	1	1	2
4	.05	--	0	1	2	3	4	4	5	6	7	8	9	10
	.01	--	--	0	0	0	1	1	2	2	3	3	4	5
5	.05	0	1	2	3	5	6	7	8	9	11	12	13	14
	.01	--	--	0	1	1	2	3	4	5	6	7	7	8
6	.05	1	2	3	5	6	8	10	11	13	14	16	17	19
	.01	--	0	1	2	3	4	5	6	7	9	10	11	12
7	.05	1	3	5	6	8	10	12	14	16	18	20	22	24
	.01	--	0	1	3	4	6	7	9	10	12	13	15	16
8	.05	2	4	6	8	10	13	15	17	19	22	24	26	29
	.01	--	1	2	4	6	7	9	11	13	15	17	18	20

U is larger than the critical values for $\alpha=0.05$ or 0.01
 → H_0 cannot be rejected
 → test non significant



Wilcoxon Signed Rank Test (2 paired probes)

- 2 samples with numerical values

$$X = \{x_1, x_2, \dots, x_n\} \quad Y = \{y_1, y_2, \dots, y_n\}$$

- D_i = differences between pairs
- R_i = ranks of the differences $|D_i|$
- Test statistics:

$$W_+ = \sum_{i=1}^{N_+} R_{i,D_i>0} \quad W_- = \sum_{i=1}^{N_-} R_{i,D_i<0}$$

$$W = \min(W_+, W_-)$$

- Question: do the positive/negative differences have different ranks?
→ H_0 : no!

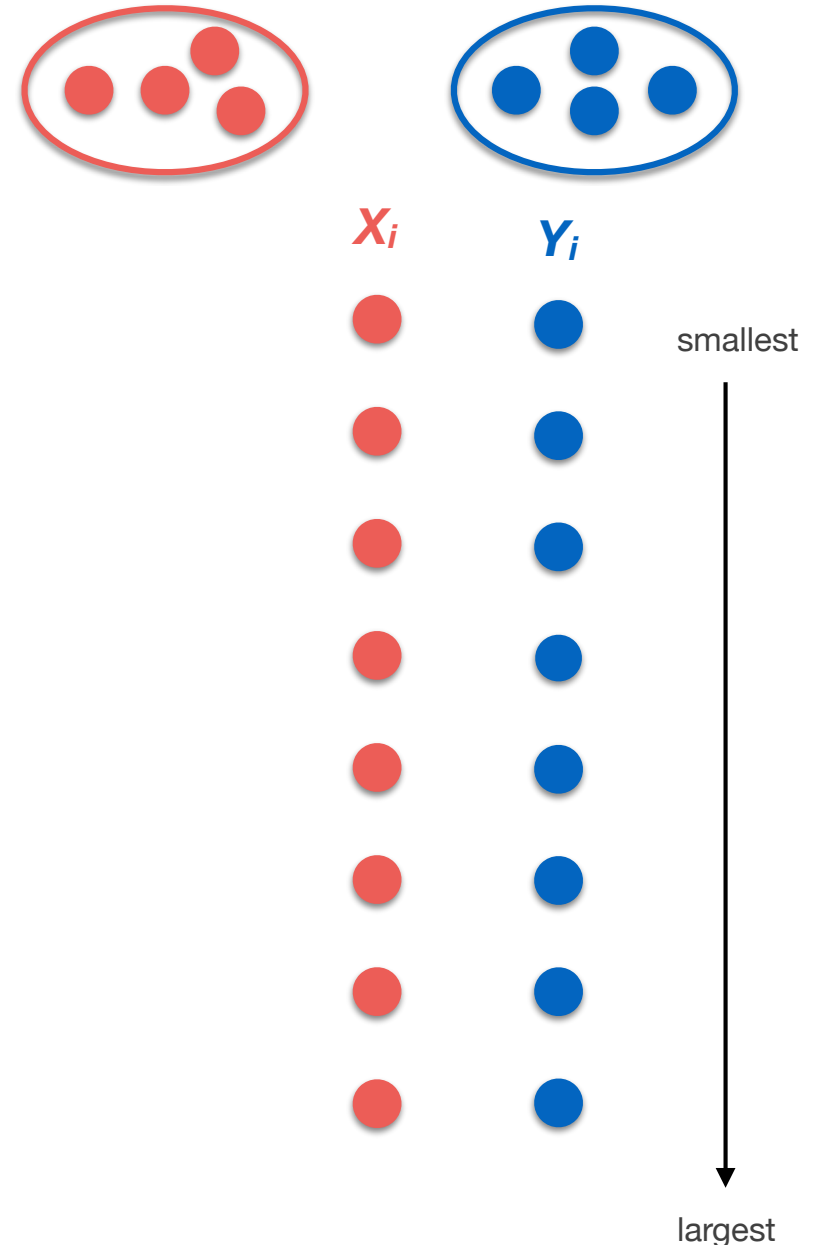


Table of critical values for Wilcoxon signed-rank test

- The smaller, the more significant
- Example:
 - $n=14$
 - $W = 22$
 - Non-significant for 2-tailed test and $\alpha = 0.05$
 - Significant for 1-tailed test and $\alpha = 0.05$
 - Non significant for 1-tailed test and $\alpha = 0.01$

n	Two-Tailed Test		One-Tailed Test	
	$\alpha = .05$	$\alpha = .01$	$\alpha = .05$	$\alpha = .01$
5	--	--	0	--
6	0	--	2	--
7	2	--	3	0
8	3	0	5	1
9	5	1	8	3
10	8	3	10	5
11	10	5	13	7
12	13	7	17	9
13	17	9	21	12
14	21	12	25	15
15	25	15	30	19
16	29	19	35	23
17	34	23	41	27
18	40	27	47	32
19	46	32	53	37
20	52	37	60	43
21	58	42	67	49
22	65	48	75	55
23	73	54	83	62
24	81	61	91	69
25	89	68	100	76
26	98	75	110	84
27	107	83	119	92
28	116	91	130	101
29	126	100	140	110
30	137	109	151	120

Wilcoxon Signed Rank Test (2 paired probes)

```
> X
  Prior Post Diff AbsDiff ranks SignedRanks
1  76.9  76.8 -0.1    0.1     1         -1
2  79.6  76.7 -2.9    2.9     2         -2
3  81.6  77.8 -3.8    3.8     3         -3
4  89.9  93.8  3.9    3.9     4          4
5  80.5  75.2 -5.3    5.3     5         -5
6  86.0  91.5  5.5    5.5     6          6
7  86.0  91.7  5.7    5.7     7          7
8  94.2 101.6  7.4    7.4     8          8
9  83.5  92.5  9.0    9.0     9          9
10 82.5  91.9  9.4    9.4    10         10
11 87.3  98.0 10.7   10.7    11         11
12 83.3  94.3 11.0   11.0    12         12
13 83.8  95.2 11.4   11.4    13         13
14 77.6  90.7 13.1   13.1    14         14
15 82.1  95.5 13.4   13.4    15         15
16 86.7 100.3 13.6   13.6    16         16
17 73.4  94.9 21.5   21.5    17         17

> W.p <- sum(X[X$Diff>0,'ranks'])
> W.m <- sum(X[X$Diff<0,'ranks'])

> W.p
[1] 142
> W.m
[1] 11
```

```
>
wilcox.test(X$Prior,X$Post,paired=TRUE)

      Wilcoxon signed rank test

data:  X$Prior and X$Post
V = 11, p-value = 0.0008392

alternative hypothesis: true location
shift is not equal to 0
```

(two-sided test)

7. Hypothesis testing

7.4 Proportion tests

Proportion tests

- This class of tests can be used when searching for
 - **relation between different categorical variables**
Is there a relation between social background and school grades?
 - comparison of **observed** vs. **expected** counts
Is there a significant gender bias in the math department if 4 professors out of 10 are women?
- Two tests are generally used
 - **Fisher-Exact test** (FET): gives an exact p-value, used for small samples
 - **chi-square test**: for larger samples ($n > 5$ in each category)
 - both tests are equivalent for large n

Fisher Exact Test

- Tests for a significant relationship between 2 variables
- Starting point: contingency table

	iPhone	other	Total
Men	4	1	5
Women	2	3	5
Total	6	4	10

Proportion iPhone/other:

- Men : $4/1 = 4$
- Women: $2/3 = 0.66$

Odds-Ratio:

$$OR = (4/1)/(2/3) = 6$$

If we would randomly distribute 6 iPhone and 4 other smartphones to 5 men and 5 women, how often would we get a larger/smaller*/more extreme** odds-ratio?

*smaller: $< 1/6$

**More extreme: > 6 or $< 1/6$

What is H₀?

	iPhone	other	Total
Men	3	2	5
Women	3	2	5
Total	6	4	10

H₀: The proportion of men with iPhone is **equal**
to the proportion of women with iPhones (2-sided)

$$OR = 1$$

H₀: The proportion of men with iPhones is **not higher**
that the proportion of women with iPhones (1-sided)

$$OR \leq 1$$

H₀: The proportion of men with iPhones is **not lower**
that the proportion of women with iPhones (1-sided)

$$OR \geq 1$$

Random permutations

If I randomly distribute 6 iPhones and 4 other phones to 5 women and 5 men, how likely it is to obtain this table?

	iPhone	other	Total
Men	4	1	5
Women	2	3	5
Total	6	4	10

Random permutations

	iPhone	other	Total
Men	4	1	5
Women	2	3	5
Total	6	4	10

$$p = \frac{\binom{6}{4} \cdot \binom{5}{4} \cdot 4! \cdot \binom{5}{2} \cdot 2! \cdot \binom{4}{1} \cdot 3!}{10!} = 0.238 \quad OR = 6$$

	iPhone	other	Total
Men	5	0	5
Women	1	4	5
Total	6	4	10

$$p = 0.023; \quad OR = \frac{5/0}{1/4} = +\infty$$

	iPhone	other	Total
Men	3	2	5
Women	3	2	5
Total	6	4	10

$$p = 0.4761; \quad OR = \frac{3/2}{3/2} = 1$$

Random permutations

	iPhone	other	Total
Men	2	3	5
Women	4	1	5
Total	6	4	10

$$p = 0.238; \quad OR = \frac{2/3}{4/1} = 1/6$$

	iPhone	other	Total
Men	1	4	5
Women	5	0	5
Total	6	4	10

$$p = 0.023; \quad OR = \frac{1/4}{5/0} = 0$$

$$p_{1-sided} = 0.238 + 0.0238 = 0.2619 \quad (OR \geq 6)$$

$$p_{2-sided} = 0.238 + 0.0238 + 0.238 + 0.0238 = 0.5238$$

$$(OR \leq \frac{1}{6} \quad \text{or} \quad OR \geq 6)$$

MoBi 2018/2019

	iPhone	other	Total
Men	8	19	27
Women	16	16	32
Total	24	35	59

Fisher's Exact Test for Count Data

data: X

p-value = 0.1831

alternative hypothesis: true odds ratio is not equal to 1

95 percent confidence interval:
0.1230632 1.3943512

sample estimates:

odds ratio
0.4273899

chi-square test

- The chi-square test compares **observed** and **expected** counts
- Starting point is a **contingency table**
- Test statistics

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}$$

- H_0 : expected and observed proportions are equal
- H_0 distribution: chi2 distribution with $n-1$ degrees of freedom for n observations
- Application possible when $O_i > 2$ and $O_i > 5$ in 80% of observations
- *Note: the chi-square test is always a 1-sided upper tail test!*

Observed

	iPhone	other	Total
Men	14	30	44
Women	5	20	25
Total	19	50	69



Observed proportions

	iPhone	other	Total
Men	31.8 %	68.2 %	100 %
Women	20 %	80 %	100 %
Total	27.5 %	72.5 %	100 %



Expected counts under H0

	iPhone	other	Total
Men	12.1	31.9	44
Women	6.9	18.1	25
Total	19	50	69



H0 proportions

	iPhone	other	Total
Men	27.5 %	72.5 %	100 %
Women	27.5 %	72.5 %	100 %
Total	27.5 %	72.5 %	100 %

$$\chi^2 = \frac{(14 - 12.1)^2}{12.1} + \frac{(30 - 31.9)^2}{31.9} + \frac{(5 - 6.9)^2}{6.9} + \frac{(20 - 18.1)^2}{18.1}$$

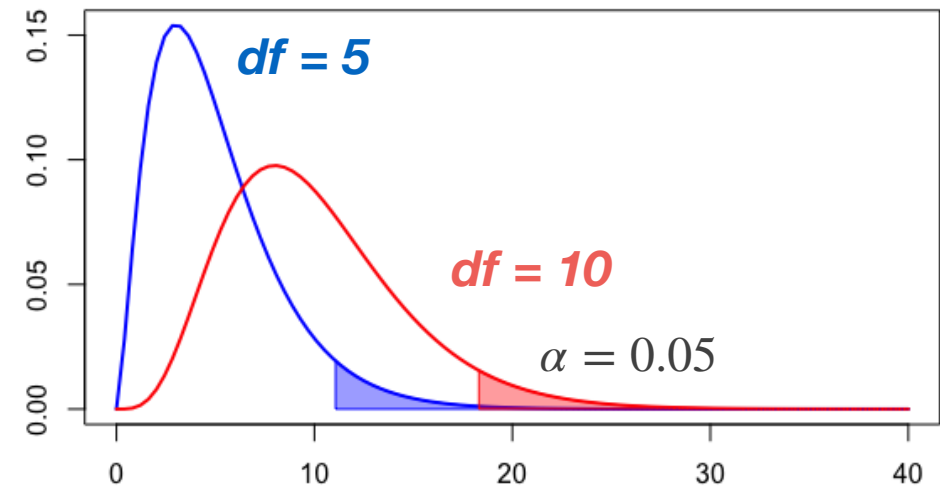
$$= 0.6022$$

degrees of freedom = (rows-1) x (columns-1)

chi-square distribution

Critical values

	0.025	0.05	0.1
df = 1	5.02	3.84	2.71
df = 2	7.38	5.99	4.61
df = 3	9.35	7.81	6.25
df = 4	11.14	9.49	7.78
df = 5	12.83	11.07	9.24
df = 6	14.45	12.59	10.64
df = 7	16.01	14.07	12.02
df = 8	17.53	15.51	13.36
df = 9	19.02	16.92	14.68
df = 10	20.48	18.31	15.99



$$\alpha = 0.05$$

$$\chi^2 = 0.6022$$

$$df = 1$$

not significant...

More than 2 categories

Side effects

	weak	medium	strong	Total
Drug A	25	11	13	49
Drug B	9	14	11	34
Total	34	25	24	83

	weak	medium	strong	Total
Drug A	51 %	22.5 %	26.5 %	100 %
Drug B	26.5 %	41.2 %	32.3 %	100 %
Total	41 %	30.1 %	28.9 %	100 %

```
> table(sideeffect)
  SideEffect
Drug weak medium strong
  A      25      11      13
  B       9      14      11

> chisq.test(table(sideeffect))
  Pearson's Chi-squared test
data:  table(sideeffect)
X-squared = 5.5257, df = 2, p-value = 0.06311

> fisher.test(table(sideeffect))
  Fisher's Exact Test for Count Data
data:  table(sideeffect)
p-value = 0.06375
alternative hypothesis: two.sided
```

8. Power of a test

Reliability of statistical test

	H_0 is valid	H_0 is NOT valid	
H_0 rejected ($p < \alpha$)	False-positive (type 1 error)	True positive	test positive
H_0 not rejected ($p > \alpha$)	True negative	False-negative (type 2 error)	test negative
	negative	positive	

- A **reliable test** should have a small number of false-positives and false-negatives
- Increasing significance level leads to **????** false-positives and **???** false-negatives

Reliability of statistical test

	H ₀ is valid	H ₀ is NOT valid	
H ₀ rejected (p < α)	FP	TP	test positive
H ₀ not rejected	TN	FN	test negative
	negative	positive	

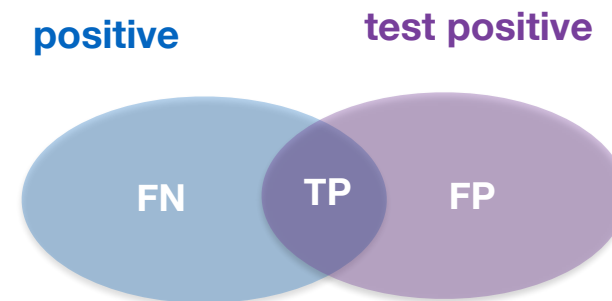
$$\text{false-negative rate (FNR)} = \frac{FN}{\text{positives}} = \frac{FN}{FN + TP}$$

$$\text{false-positive rate (FPR)} = \frac{FP}{\text{negatives}} = \frac{FP}{FP + TN}$$

$$\text{false-discovery rate (FDR)} = \frac{FP}{\text{test positives}} = \frac{FP}{FP + TP}$$

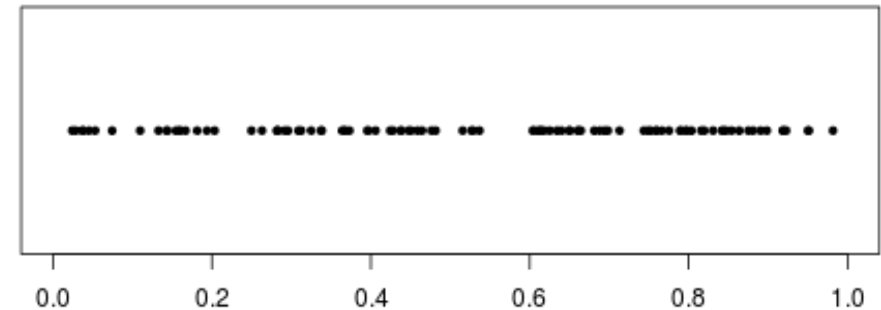
$$\text{precision} = \frac{TP}{\text{test positives}} = \frac{TP}{FP + TP}$$

$$\text{recall} = \frac{TP}{\text{positives}} = \frac{TP}{FN + TP}$$

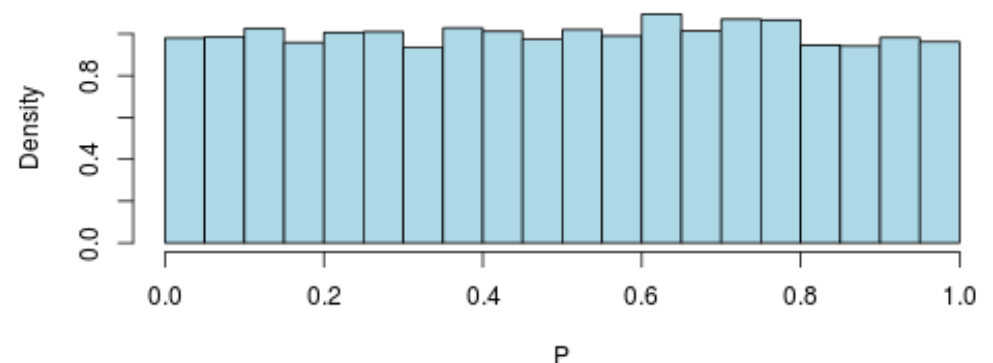


P-value distribution under H_0

- *What are typical p-values under H_0 ?*
- **Experiment:** draw 2 sets (S_1 & S_2) of 50 random numbers each **from the same distribution**
- *H_0 : the expectation of both distributions are equal (TRUE!)*
- Compute t-test between S_1 and S_2 , and determine P-value
- Repeat this experiment 1000 times, and plot the distribution of the 1000 p-values



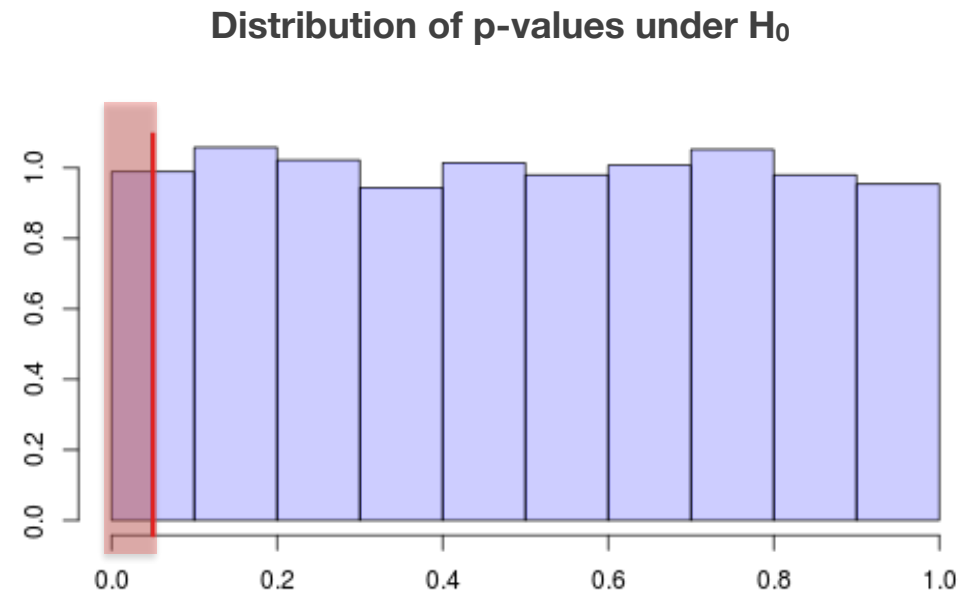
Verteilung der P-Werte bei H_0



Distribution of p-values under H_0 = uniform distribution

Type 1 errors

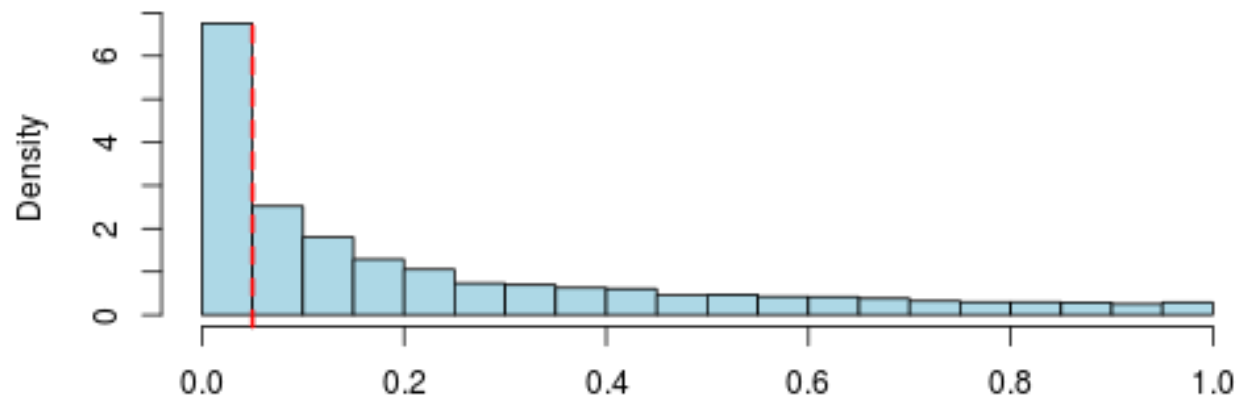
- **Red area:**
with $\alpha = 5\%$, we would have
wrongly rejected H_0
→ **FALSE POSITIVE**
- **How often would that occur?**
→ red area compared to the total
area = 5% because uniform
distribution



α is the **FALSE-POSITIVE RATE (FPR)**

P-value distribution under H_1

- Experiment: draw 2 sets (S_1 & S_2) of 50 random numbers each **from two distributions with different expectation**
- *H_0 : the expectation of both distributions are equal (FALSE!)*
- compute p-value using a 2 sample t-test
- Repeat 1000 times and plot distribution of p-values



Many small p-values
→ H_0 would have been rejected

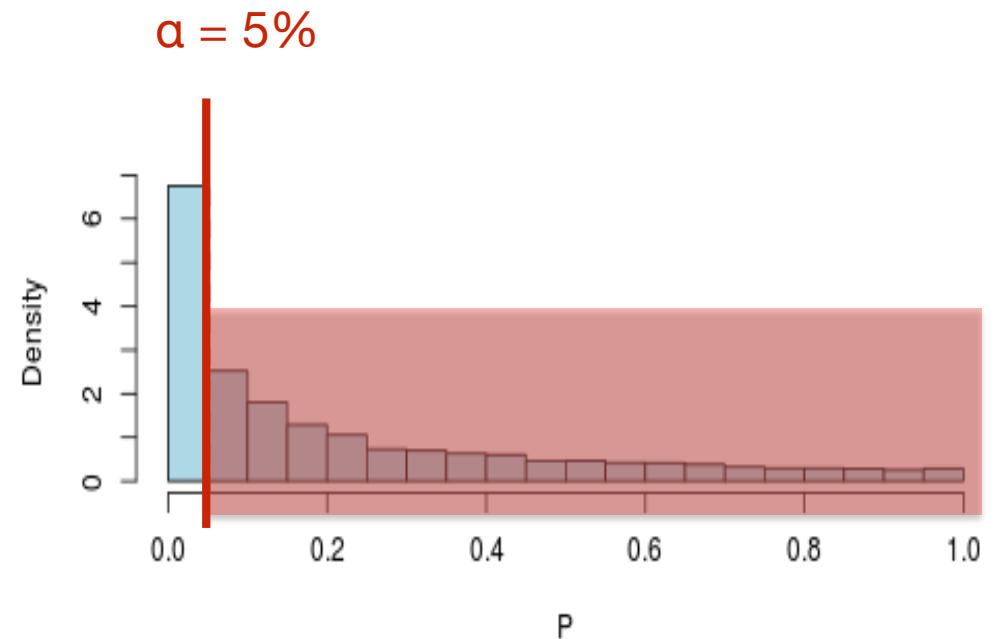


Some large p-values
→ H_0 would have NOT been rejected



Type 2 errors

- Occur when a false H_0 hypothesis is **NOT rejected** by the test
→ False-negative (Type 2 errors)
- Probability of a type 2 error:
 β - value
- Probability for a type 2 error NOT to occur
→ **power of a test = $1 - \beta$**

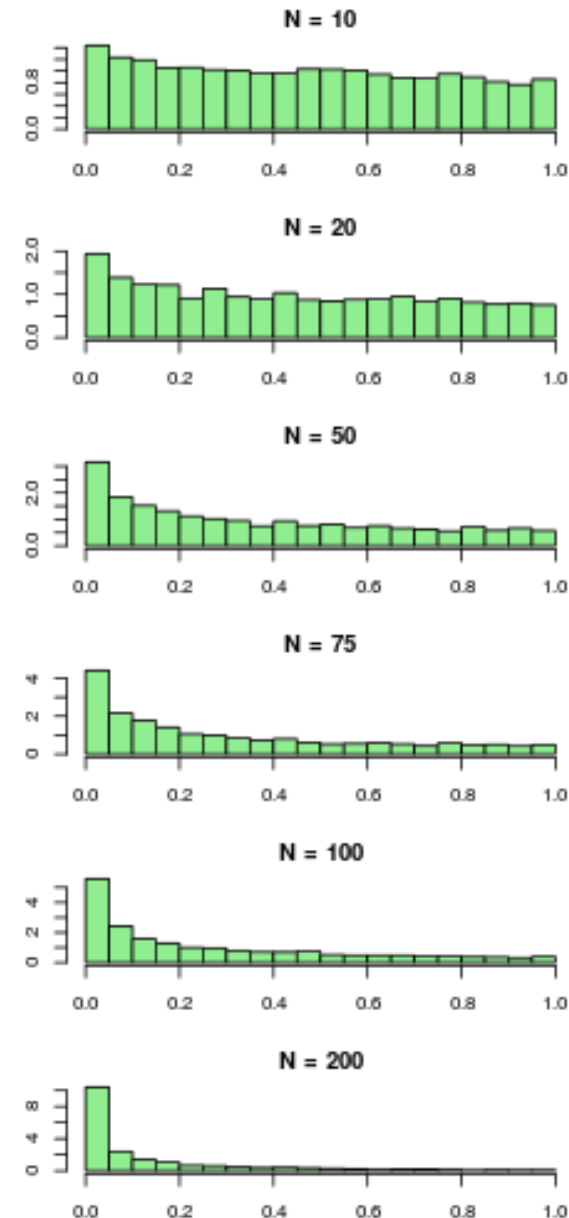


This area represents the cases for which H_0 will not be rejected
→ **false-negatives**

Power of a test

- Generate 2 datasets of length n
 - one from a normal distribution with mean 0
 - one from a normal distribution with mean 0.2
- **H_0 : expectation of both underlying distributions is identical (False!)**
- perform t-test, compute p-values for various values of n

$$\beta \xrightarrow{n \rightarrow \infty} 0$$



Power of a test

- The power depends on:
 - **Significance level α**
 - **Sample size n**
 - **Effect-size:** how strong is the observed effect?

Power

large α

small α

large
sample size

small
sample size

large
effect size

small
effect size